SMART: Selective Marker-guided Adaptive Recursive Transformer for Transcriptomic Classification

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Abstract

Transformer foundation models for transcriptomics treat every one of the ∼20,000 measured genes as an equally important token and stack many independent layers, which makes them parameter-heavy and computationally prohibitive. We argue that, for gene expression, *parameter efficiency should be an architectural property rather than something recovered by post-hoc pruning*. We introduce **SMART** (Selective Marker-guided Adaptive Recursive Transformer), a transformer that (i) learns end-to-end which genes are *marker genes* worthy of dedicated computation, using a cross-attention *marker router* in which learnable marker queries attend over all genes, so gradients reach every gene rather than a frozen pre-selected set; (ii) represents each sample by only its $M \ll N$ selected markers, reducing self-attention cost from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$; and (iii) processes these marker tokens with a *single* transformer block applied recursively, where a Mixture-of-Recursions router grants each gene its own *adaptive* recursion depth. Most genes exit after one pass while a few disease drivers are iterated deeper, so a gene's recursion depth becomes an intrinsic, compute-allocation based importance score instead of a post-hoc attention map. On pan-cancer cohort classification we reach 98.5% accuracy and 98.7% macro-F1 while using 3.99× fewer transformer-stack parameters than an equivalent stack of independent layers and only 0.60× of the fixed-depth recursion FLOPs. Beyond cohort identity we evaluate SMART on five harder clinical phenotype tasks defined on the same tumours (stage, tumour T, node N, overall survival, tumour status): under an identical split and the full gene set it attains the best mean macro-F1 against ten strong nonlinear baselines, including XGBoost, LightGBM and CatBoost, and it transfers unchanged to four single-cell cell-type datasets. A controlled selection study shows the learned selectors beat random and variance selection when a marker signal is present, and the per-gene recursion depth yields an interpretable, compute-allocated gene panel; multi-seed ablations report, transparently, that on the weakly-determined clinical labels the architectural choices are statistically within noise. We further make routing *biology-informed* by injecting a label-free genomap gene-gene interaction prior into the recursion router, and test it against a degree-matched random-graph control; on these datasets it does not separate from that control, a negative result we report transparently alongside its mathematical and biological foundation. The whole pipeline, including training, ablations, baselines, and this paper itself, regenerates from a single shell script.

1 Introduction

Single-cell and bulk RNA sequencing now routinely profile the expression of tens of thousands of genes across millions of cells, and a wave of transformer-based *foundation models* such as scGPT (Cui et al. 2024), Geneformer (Theodoris et al. 2023), scBERT (Yang et al. 2022) and scFoundation (Hao et al. 2024) has adapted the architecture of (Vaswani et al. 2017) to this modality. These models are powerful, but they inherit two costly habits from natural-language transformers. First, they treat *every* gene as an equally important token, so a housekeeping gene and a lineage-defining marker receive identical computational budgets. Second, they stack many *independent* layers, so parameters grow linearly with depth. The result is models with tens to hundreds of millions of parameters (Hao et al. 2024) whose self-attention scales quadratically in the number of genes, and whose efficiency is usually addressed only afterwards through pruning or distillation.

We take a different stance: *for gene expression, the data themselves tell us where to spend computation*. Decades of single-cell biology rest on the observation that a small set of *marker genes* is sufficient to discriminate cell types and disease states (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023). If a model could decide, while training, which genes are markers, it could grant them dedicated capacity and let everything else share parameters. This makes parameter efficiency an *architectural* property rather than a compression afterthought.

We realise this idea in **SMART** (Selective Marker-guided Adaptive Recursive Transformer), a recursive marker-guided transformer with three coupled components. (1) A cross-attention *marker router*: M learnable marker queries attend over all N genes (Set-Transformer / Perceiver-style (Jang, Gu, and Poole 2017; Balın, Abid, and Zou 2019)), with a temperature-annealed softmax over genes, so the model learns *which* genes are markers end-to-end while gradients reach every gene. This matters in practice: we show that hard top-k routing cannot explore and underperforms even random panels. (2) *Marker-driven compression*: each sample is represented by only its $M \ll N$ selected markers, cutting attention from $\mathcal{O}(N^2)$ to $\mathcal{O}(M^2)$. (3) A *recursive shared block*: instead of K independent layers we apply one transformer block K times, in the spirit of Universal Transformers (Dehghani et al. 2019), ALBERT (Lan et al. 2020) and the recent Mixture-of-Recursions (Bae et al. 2025). After each recursion we re-run a marker gate on the updated representations, so the model jointly learns *what biological information to preserve* and *how to allocate its computational capacity*.

We make the following contributions:

- We propose SMART, to our knowledge the first transformer for transcriptomics in which marker selection, token compression, and parameter-shared recursion are co-designed and trained end-to-end.
- We introduce *recursive marker refinement*, a closed feedback loop that turns marker identification from a preprocessing step into a differentiable part of the architecture.
- We adapt Mixture-of-Recursions (Bae et al. 2025) to gene expression: an expert-choice router gives each gene an *adaptive* recursion depth, unifying weight sharing with adaptive computation and turning a gene's recursion depth into an intrinsic importance score that needs no post-hoc attribution.
- We show, on pan-cancer cohort classification, that the model matches strong accuracy while using 3.99× fewer transformer parameters than independent layers, and that learned markers beat random and variance baselines, with the recursion-depth signal yielding an interpretable gene panel.
- We benchmark SMART beyond cohort identity on five harder clinical phenotype tasks (stage, tumour T, node N, overall survival, tumour status) against ten strong nonlinear baselines, including gradient-boosted trees (XGBoost, LightGBM, CatBoost), where SMART attains the best across-task mean macro-F1, and we transfer it unchanged to four single-cell datasets.
- We report multi-seed ablations and depth / marker-count sweeps, and state honestly that on the weakly-determined clinical labels the architectural choices are within noise: the marker-learning advantage requires a genuine marker signal in the label, as on the cohort task.
- We propose a *biology-informed router* that injects a label-free genomap gene-gene interaction prior (network centrality) into the recursion router as an annealed additive bias, and we evaluate it against a degree-matched random-graph control; on the present datasets it does not separate from that control, a negative result we report transparently alongside the mechanism and its theory.
- We release a fully reproducible pipeline in which a single shell script runs all experiments, baselines, and regenerates this paper, numbers and tables included.

2 Related Work

Transformer foundation models for omics. Geneformer (Theodoris et al. 2023) and scBERT (Yang et al. 2022) adapt masked-language-model pretraining to single-cell transcriptomes; scGPT (Cui et al. 2024) scales generative pretraining to 33M cells; scFoundation (Hao et al. 2024) trains a 100M-parameter model over ∼20,000 genes. Closest to our setting, GexBERT (Jiang and Hassanpour 2025) pretrains a gene-plus-value bulk RNA-seq transformer autoencoder with a masking-and-restoration objective and applies it to pan-cancer classification and survival; it shares our gene-plus-value embedding but, like the others, treats genes uniformly and uses independent layers, with no marker-driven sparsity or recursion. Earlier deep generative approaches such as scVI (Lopez et al. 2018) established probabilistic latent representations of expression. More recent foundation models push to whole-genome transcription (Fu et al. 2025), gene-regulatory-network-aware single-cell prediction (Zhang et al. 2025), and other organisms (Cao et al. 2025). genomap (Islam and Xing 2023) instead reshapes the gene vector into an image via an optimal-transport layout for convolutional analysis; image-based CNN classifiers of TCGA RNA-seq follow the same recipe (Khalifa et al. 2020; Rukhsar et al. 2022). A parallel line improves accuracy mainly through gene selection or augmentation before a deep net (Polepalli 2025; Kim and Jang 2025; Rahaman et al. 2025; Shukla, Dwivedi, and Mishra 2025; Bouazza 2025). All of these treat genes uniformly or select them in a separate stage; none make marker-driven sparsity an architectural prior.

Parameter-efficient and recursive transformers. Tying weights across depth was shown to retain representational power by Universal Transformers (Dehghani et al. 2019) and to drastically shrink models in ALBERT (Lan et al. 2020). Mixture-of-Recursions (Bae et al. 2025) unifies weight sharing with token-level adaptive depth, and Mixture-of-Depths (Raposo et al. 2024) routes tokens through variable numbers of layers, both echoing sparsely-gated mixtures of experts (Shazeer et al. 2017) and adaptive computation time (Graves 2016). A separate line of work on efficient attention, including Linformer (Wang et al. 2020), Performer (Choromanski et al. 2021) and Nyströmformer (Xiong et al. 2021), reduces the quadratic cost generically. Recursive weight sharing has also been pushed in vision and restoration transformers, where a shared block is unrolled with depth, sometimes with input-conditioned modulation: the Sliced Recursive Transformer (Shen, Liu, and Xing 2022), RISTRA (Zhou et al. 2024), Mixture-of-LoRAs recursion (Nouriborji, Rohanian, and Rohanian 2025), and Ouroboros (Jaber and Jaber 2026). We borrow the weight-sharing mechanism but make the token set itself biologically structured, so our $\mathcal{O}(M^2)$ saving is complementary to these methods.

Markers, pathways, and biological priors. Marker-based annotation tools such as scType (Ianevski, Giri, and Aittokallio 2022) and SingleR (Aran et al. 2019), and curated databases including PanglaoDB (Franzén, Gan, and

Björkegren 2019) and CellMarker 2.0 (Hu et al. 2023), encode the long-standing principle that few genes carry most discriminative signal. Pathway-informed models such as the recent graph transformer PATH (Howlader, Islam, and Le 2026) build structure from Reactome (Gillespie et al. 2022). We use such resources only to *validate* our learned markers, keeping marker discovery fully data-driven. Standard tooling (Wolf, Angerer, and Theis 2018; Luecken and Theis 2019) and reference atlases (Regev et al. 2017; The Tabula Sapiens Consortium 2022) provide the broader context for our task.

3 Method

Overview

Let $x \in \mathbb{R}^N$ be the expression vector of a sample over N genes. SMART maps x to class logits through five stages: gene embedding, learnable marker selection, marker-anchored compression, recursive shared transformation with marker refinement, and a pooled classifier. Figure 1 summarises the two stages.

Gene Embedding

Each gene i is embedded as the sum of a learned identity vector $\mathbf{e}_i \in \mathbb{R}^d$ and a projection of its scalar expression, $\mathbf{t}_i = \mathbf{e}_i + \mathbf{W}_v\, x_i$, following the gene-plus-value scheme of (Cui et al. 2024; Theodoris et al. 2023). This is linear in N and so runs over all genes before any compression.

Cross-Attention Marker Router

We identify markers with a cross-attention *router*, the best-practice form of differentiable selection (Set-Transformer induced points and Perceiver-style cross-attention). We maintain M learnable marker queries $\mathbf{q}_m \in \mathbb{R}^d$; each attends over the genes through a shared key projection $\mathbf{k}_i = \mathbf{W}_k\mathbf{e}_i$, giving selection weights $\mathbf{w}_m = \mathrm{softmax}\big(\mathbf{q}_m\mathbf{K}^\top/(\tau\sqrt{d})\big)$ over *all* N genes, with temperature τ annealed from soft to peaked. Because the softmax spans all genes, gradients reach every gene, so a query can migrate to an informative gene it did not initially favour, which is exactly the property that hard top-k routing lacks. Two ingredients are essential: the all-gene softmax, and a *peaked initialisation* that points each query at a distinct gene's key, so training starts at random-selection quality instead of a uniform average of all genes. At inference each query collapses to its arg-max gene $g_m = \arg\max_i \mathbf{q}_m^\top\mathbf{k}_i$, giving discrete, interpretable markers and $\mathcal{O}(M^2 d)$ attention. As alternatives we consider the closely related *Concrete* selector (Balın, Abid, and Zou 2019; Jang, Gu, and Poole 2017) (free per-slot logits over genes), fixed *variance*- and *random*-selected panels, and a naive hard *router* in the style of expert-choice routing (Shazeer et al. 2017; Raposo et al. 2024; Bae et al. 2025); the last fails because hard routing cannot explore.

Marker Tokens

Each query produces one marker token combining the (soft-)selected gene identity and its expression, $\mathbf{c}_m = (\mathbf{w}_m^\top\mathbf{E}) + \mathbf{W}_v\,(\mathbf{w}_m^\top\mathbf{x})$, where $\mathbf{E} \in \mathbb{R}^{N\times d}$ are the gene-identity embeddings. The identity and value contributions are summed without an intermediate LayerNorm; the two are placed on a common scale by the pre-norm LayerNorm at the entry of the shared block (Sec. 3), which normalises every marker token before its first self-attention. (In the optional aggregation variant the per-gene tokens are LayerNorm-ed during embedding before pooling.) As an optional *aggregation* variant, non-selected genes can instead be folded into their nearest marker by cosine similarity; we find this makes the model robust to marker choice but masks selection quality, so our headline model uses pure selection.

Recursive Shared Transformer with Marker Refinement

Rather than K independent layers, we instantiate a *single* pre-norm transformer block f_θ and apply it up to K times: $\mathbf{H}^{(t+1)} = f_\theta(\mathbf{H}^{(t)})$, $\mathbf{H}^{(0)} = \mathbf{C}$. This ties all depth-wise parameters (Dehghani et al. 2019; Lan et al. 2020; Bae et al. 2025), so the stack's parameter count is independent of K. After each pass we recompute a per-token gate from the *updated* cluster embeddings, $g_m^{(t)} = \sigma(\mathrm{MLP}_r(\mathbf{H}_m^{(t)}))$, and apply it before the next pass. We call this *recursive marker refinement*; it lets markers that remain informative survive while others are suppressed.

Mixture-of-Recursions over genes. Spending the full depth K on every marker is wasteful: most genes are settled after one pass, while a few disease drivers reward deeper iterative reasoning. We therefore make the recursion *adaptive per token* with a Mixture-of-Recursions router (Bae et al. 2025) (Figure 2), turning weight-shared recursion (parameter efficiency) and adaptive computation (Graves 2016; Raposo et al. 2024) into a single co-designed mechanism. Our headline model uses *expert-choice* routing: at step t a lightweight router scores the active marker tokens and a capacity c_t (a geometric funnel $1, \frac{1}{2}, \frac{1}{4}, \ldots$) keeps only the top-$\lceil c_t M\rceil$; selected tokens are gated by the router weight and updated by f_θ, the rest are frozen at their current state. Survivors at step t are the only candidates at step $t{+}1$, so genes are progressively filtered and the tokens that reach the deepest step are, by construction, those the model judges most worth computing on. We define a gene's *recursion depth* $d_m \in \{0, \ldots, K\}$ as the number of steps its marker survived; averaged over a cohort this is an intrinsic, compute-allocation based importance score, a biomarker signal read directly off the architecture rather than from post-hoc attention. As an ablation we also implement *token-choice* routing (Bae et al. 2025), where each gene selects a single depth up front (top-1 over $\{1, \ldots, K\}$) with a Switch-style load-balancing loss (Shazeer et al. 2017); consistent with the original report we find expert-choice the stronger of the two. Both share the block f_θ, so the parameter-efficiency claim is untouched, and both reduce to the uniform fixed-depth recursion when routing is disabled. The router is trained with a small logit z-loss (and, for token-choice, the balancing term) added to the objective.

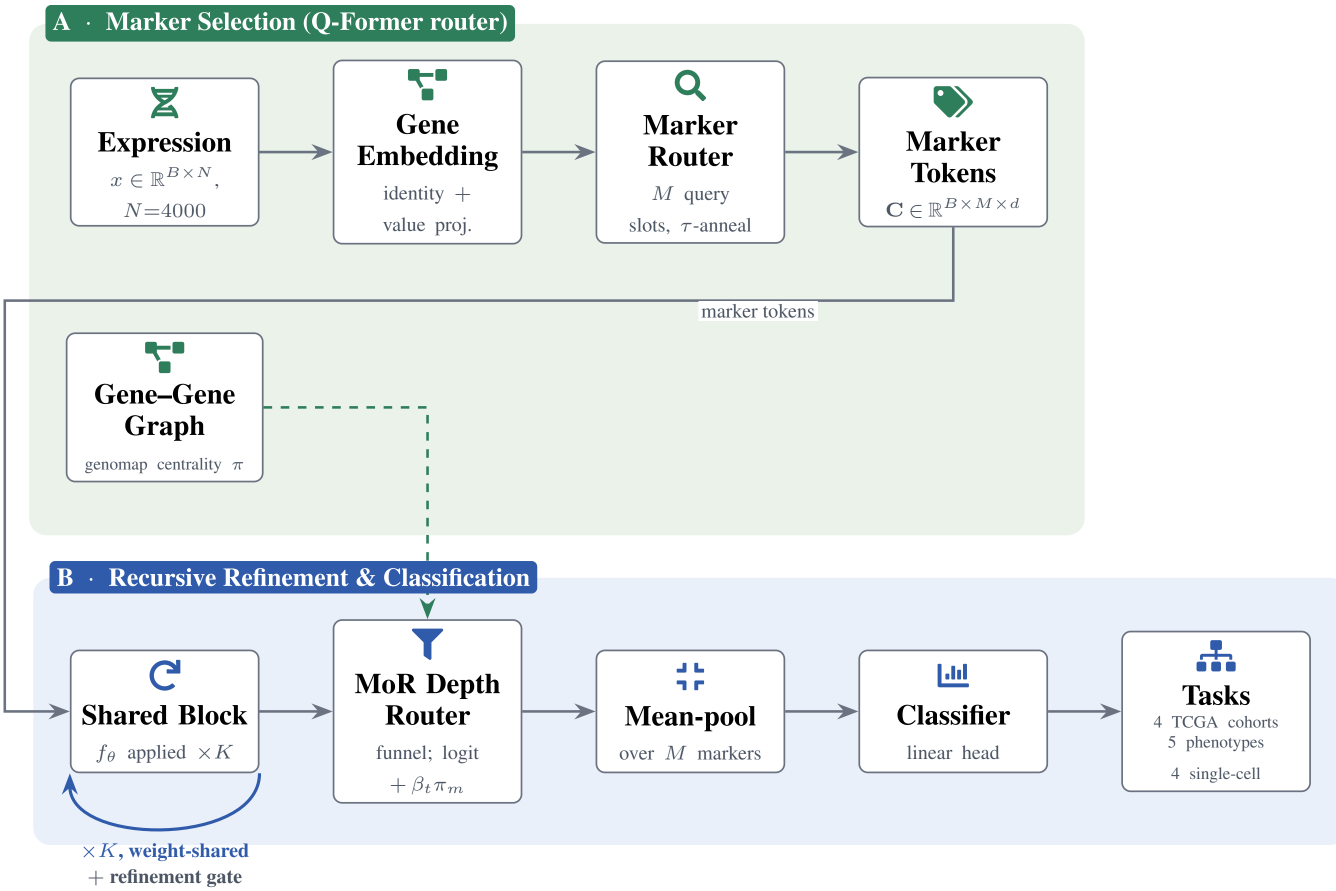

Figure 1: **System overview. Panel A (marker selection):** the expression vector is embedded gene-by-gene, then M learnable query slots cross-attend over *all* N genes (temperature annealed soft→peaked) to select interpretable marker tokens, a Q-Former-style router. **Panel B (recursive refinement & classification):** a *single* weight-shared block f_θ is applied up to K times (loop-back arrow) with a per-marker refinement gate between passes; a Mixture-of-Recursions router funnels capacity so each marker gets an *adaptive* depth d_m (deeply routed genes are candidate drivers). **Biology-informed routing:** a genomap gene-gene co-expression graph supplies a network-centrality prior π that is added (annealed by β_t) to the depth-router logit (dashed arrow), so co-expression hub genes get a head start in the funnel without any label leakage. Tokens are mean-pooled and classified; the *same* pipeline serves all datasets, the four TCGA cohorts, five clinical phenotype tasks, and four single-cell cell-type datasets.

Biology-Informed Routing

So far the router decides depth from data alone, and biology only enters afterwards, when we cross-check the deepest-routed genes against curated markers. We instead move a biological prior *into* the routing decision. For marker token m at step t, the expert-choice logit becomes

$$\tilde{r}_m^{(t)} = \underbrace{\tfrac{1}{\tau}\,\mathbf{w}_r^\top \mathbf{H}_m^{(t)}}_{\text{data-driven (learned)}} + \underbrace{\beta_t\,\pi_m}_{\text{biological prior}}, \tag{1}$$

and the keep/drop top-$\lceil c_t M\rceil$ and the gate $g_m^{(t)} = \sigma(\tilde{r}_m^{(t)})$ run exactly as before, so the recursion-depth definition is unchanged but now reflects prior and data together. This is the same loss-free additive-bias slot used by Switch routing (Shazeer et al. 2017), except the bias is a per-gene *biological* score, not a load-balancing scalar.

The prior π_m from gene-gene interactions. We obtain π_m from genomap's gene-gene *interaction* identification (Islam and Xing 2023): genomap's `createInteractionMatrix` defines interaction as the pairwise correlation distance between genes across samples (which it then feeds to optimal transport for an image layout). We call that function on the training split and reuse only its interaction matrix, taking the co-expression affinity $\mathbf{W}_{ij} = |\mathrm{corr}(g_i, g_j)| = |1 - d_{ij}|$ from genomap's distance d_{ij}, sparsifying it to each gene's k nearest neighbours, symmetrising it, and reading off the network centrality

$$\pi = \mathrm{zscore}\big(\text{eigvec-centrality}(\mathbf{W})\big), \tag{2}$$

so co-expression *hub* genes, the master-regulator-like nodes whose perturbation propagates widely, receive a larger prior and are nudged to recurse deeper. Crucially $\mathbf{W}$ is built from

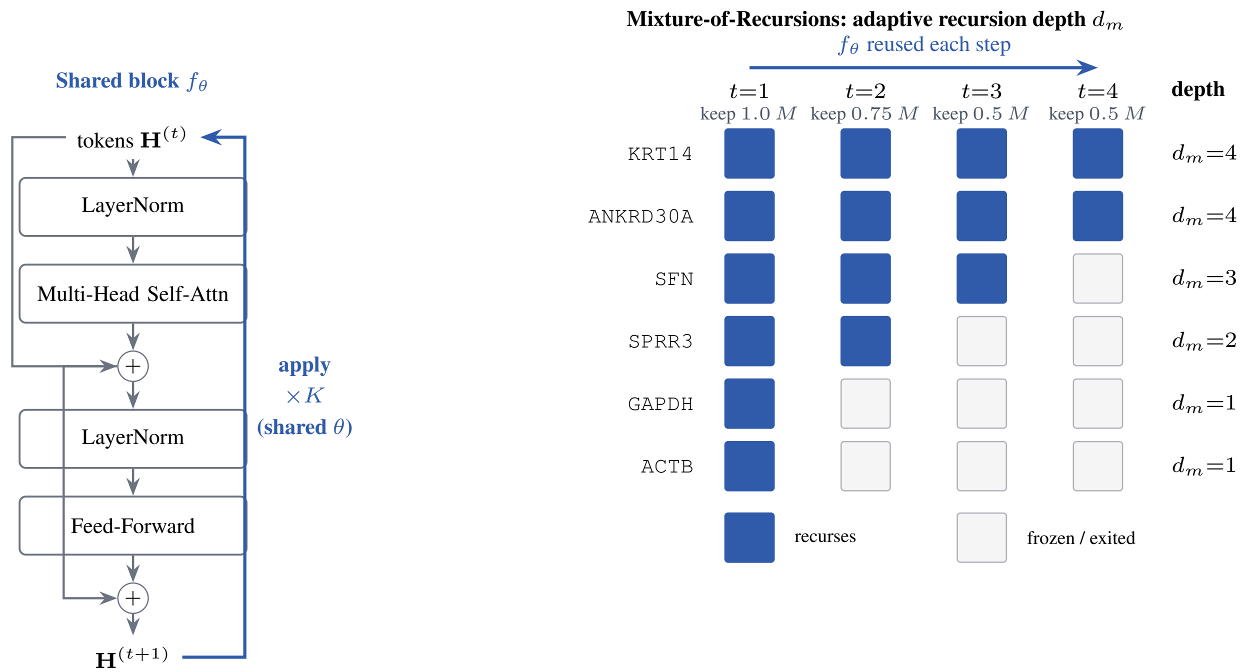

Figure 2: **Adopting Mixture-of-Recursions.** *Left:* one weight-shared pre-norm block f_θ (the model's only transformer parameters) is re-applied up to K=4 times (recurrence arrow), so depth costs no extra parameters. *Right:* an expert-choice router keeps a shrinking top fraction of markers per step (capacity funnel $1, \frac{3}{4}, \frac{1}{2}, \frac{1}{2}$); a marker not kept is frozen, so its *recursion depth* d_m is the deepest step it survived. Driver genes (`KRT14`, `ANKRD30A`) recur deepest; settled house-keeping genes (`GAPDH`, `ACTB`) exit at d_m=1, saving compute.

expression alone, with no labels, so the prior injects biological network structure without leaking the cohort labels; this is what keeps the gene-discovery claim honest.

Annealing β_t. A fixed strong prior would behave like hard routing, pinning compute to known hubs and unable to discover new genes, exactly the failure mode of our uniform-init router. We therefore warm-start: $\beta_t = \beta_0(1 - \text{progress})$ decays to 0 over training, so the prior dominates early, when hidden states are still random and biology gives a sensible starting allocation, and the data-driven term takes over late. This is an empirical-Bayes shrinkage / curriculum: a strong prior when evidence is weak, fading as evidence accumulates. Because π_m is a constant additive bias, $\tilde{r}_m^{(t)}$ stays smooth in $\mathbf{w}_r$ and the gate $g_m^{(t)}$ still carries gradient, so nothing about trainability or the parameter-efficiency claim changes. We validate the component against a degree-matched *random*-graph control: only if the real co-expression graph beats the random one is it the biology, not mere smoothing, that helps (Sec. 4).

Training Objective

We minimise a composite loss $\mathcal{L} = \mathcal{L}_{\text{task}} + \lambda\mathcal{L}_{\text{marker}} + \gamma\mathcal{L}_{\text{div}} + \beta\mathcal{L}_{\text{comp}} + \zeta\mathcal{L}_z + \eta\mathcal{L}_{\text{bal}}$, where $\mathcal{L}_{\text{task}}$ is class-weighted cross-entropy (inverse-frequency weights on the train split); $\mathcal{L}_{\text{marker}}$ is the cross-entropy of an auxiliary linear classifier fed only the (pre-recursion) cluster tokens, which forces the marker head to select task-sufficient genes; $\mathcal{L}_{\text{div}} = \frac{1}{M(M-1)}\sum_{i\neq j}(\tilde{\mathbf{e}}_i^\top\tilde{\mathbf{e}}_j)^2$ is the off-diagonal energy of the normalised marker Gram matrix (prevents marker collapse); $\mathcal{L}_{\text{comp}} = \frac{1}{N}\sum_i \sigma(s_i)$ encourages a sparse importance distribution, where s_i is the per-gene importance logit emitted by the marker-scoring head and σ the logistic function, so the term penalises the mean selection mass and pushes most genes toward zero importance; its gradient flows only into that head. For the soft selectors used in the headline model (cross-attention router, Concrete) the temperature-annealed softmax already yields a peaked, near-one-hot selection, so this explicit sparsity term is switched off (β=0) and s_i refers to the learnable-head variant. $\mathcal{L}_z = \mathbb{E}[(\operatorname{logsumexp}\mathbf{r})^2]$ is the router logit z-loss; and $\mathcal{L}_{\text{bal}}$ is the Switch-style load-balancing term (Shazeer et al. 2017) for token-choice routing ($\mathcal{L}_{\text{bal}}$=$K\sum_i P_i f_i$, the product of mean routing probability P_i and dispatch fraction f_i at depth i; expert-choice is balanced by construction so η has no effect there). Unless noted we use λ=0.1, γ=0.05, β=0.01, ζ=10^{-3}, η=0.1; the selection temperature τ is annealed geometrically from 1 to 0.1 over training and each router query is peak-initialised on a distinct gene.

4 Experiments

Setup

We evaluate on pan-cancer bulk transcriptomes assembled from four TCGA cohorts (Weinstein et al. 2013) (breast,

Head	Accuracy	Macro-F1	Weighted-F1	#Classes
cancer type†	98.5	98.7	98.5	4

Table 1: Main results per output head († primary head). All numbers are produced directly by our pipeline.

Cohort (class)	Precision	Recall	F1	Support
Breast (BRCA)	98.9	97.8	98.4	183
Head-neck (HNSC)	97.7	100.0	98.8	85
Lung (LUNG)	97.6	97.6	97.6	169
Thyroid (THCA)	100.0	100.0	100.0	86

Table 2: Per-class (per-cohort) breakdown of the primary cancer-type head: precision, recall, F1 and test-set support for each of the four TCGA cohorts (the macro and support-weighted aggregates are in Table 1). Reported per cohort rather than as a single macro number so cohort-level strengths and weaknesses are visible.

lung, head-and-neck, thyroid), with per-gene z-scoring fit on the training split and the top 4000 high-variance genes retained. Unless noted, d=128, M=256 markers, recursion depth K=4, trained for 25 epochs with AdamW and early stopping on validation macro-F1. The primary task is cell-of-origin / cancer-type classification; the framework is data-agnostic and applies unchanged to single-cell cell-type labels. For the harder clinical phenotype tasks and the external-baseline comparison we instead use the *full* 20,530-gene vector (no high-variance pre-filter) on the unified 2,738-sample table (Appendix A); there SMART's hyperparameters are selected per task by validation macro-F1 and every baseline uses the identical split and gene set. All reported metrics use the *hard* arg-max marker panel at inference (each query collapses to its single top gene); the soft all-gene selection is used only during training, so every number reflects the discrete, interpretable panel rather than a soft mixture.

Main Results

The primary task is four-way pan-cancer cohort classification (Breast/BRCA, Head-neck/HNSC, Lung/LUNG, Thyroid/THCA), so the four classes *are* the four datasets. Since a macro average can hide per-cohort weaknesses, Table 2 reports *per-class (per-cohort)* precision, recall and F1 with support, and Table 1 the per-head aggregates. SMART attains 98.5% accuracy and 98.7% macro-F1, well balanced across cohorts (Thyroid near-perfect, the smaller Head-neck hardest). To show the result is not a single-seed artefact, we repeat the headline configuration over 5 random seeds: it attains 96.7 ± 1.4% macro-F1 and 97.0 ± 1.4% accuracy (mean $\pm$ std), so the cohort result is stable to initialisation.

Parameter Efficiency Is Architectural

Table 3 contrasts the transformer-stack parameters of our shared recursion against an equivalent stack of independent layers, across depth. The saving is exact and present *before any training*: at K=4 the shared model uses 3.99× fewer stack parameters, and the gap widens linearly with depth. The saving is built into the architecture, not recovered by pruning.

Depth K	Shared (ours)	Independent	Reduction
1	132,608	132,608	1.00×
2	132,736	265,216	2.00×
4	132,992	530,432	3.99×
6	133,248	795,648	5.97×
8	133,504	1,060,864	7.95×

Table 3: Transformer-stack parameters: shared recursion (ours) vs. independent layers. Reduction grows linearly with depth K.

Config	Stack params	Total params	Train (s)
Expert-choice (headline)	132,992	712,457	43
Fixed depth	132,480	711,945	37
Single pass (K=1)	132,608	712,073	28
Independent layers	530,432	1,109,897	37

Table 4: Total cost per configuration: transformer-stack parameters, total parameters (including the gene-identity embedding and classifier), and training wall-clock. The shared embedding dominates the non-stack parameters and is common to all variants.

To report the claim end-to-end rather than for the stack alone, Table 4 lists the total parameter count, including the gene-identity embedding and classifier, and the training wall-clock for the headline and key ablation configurations. The full headline model has 712,457 parameters and trains in 43 s on a single GPU. The gene-identity embedding (Nd parameters) dominates the non-stack count and is shared by every variant, so removing weight sharing (the "Independent layers" row) inflates the *total* far less than the 3.99× stack-only figure suggests; the stack comparison in Table 3 is therefore the meaningful axis for the architectural claim, while the total and timing are given here for completeness.

Token Reduction Validation

Parameter and FLOP savings both follow from one upstream effect: the marker router collapses the input token space. We validate this reduction quantitatively and qualitatively on the headline cohort model. The router keeps 256 marker tokens out of 4,000 gene tokens, a 93.6% reduction that lowers the self-attention cost by 244.1× ($O(N^2) \to O(M^2)$; Table 5). Reduction alone is not enough: the retained tokens must be the informative ones. Ranking every gene by its marker-query affinity, the kept tokens carry 16.3 of the total importance mass and have a mean importance of 34.13 versus 11.96 for the discarded tokens, so the router keeps disproportionately high-scoring genes rather than pruning at random. The selection importance and the compute a gene is later allocated are, however, nearly uncorrelated in rank (Spearman ρ=-0.062 between marker

Quantity	Value
Original gene tokens (N)	4,000
Retained marker tokens (M)	256
Removed tokens	3,744 (93.6%)
Attention-cost factor $(N/M)^2$	244.1×
Importance mass retained	16.3
Mean importance, kept vs. dropped	34.13 vs. 11.96
ρ(importance, recursion depth)	-0.062
Selection method	cross-attention marker router
Top-k / selection rule	M=256 slots, hard arg-max at eval

Table 5: Token Reduction Validation on the cohort task: how far the marker router shrinks the 4,000-gene input, whether the kept tokens are the informative ones (importance mass and kept-vs-dropped importance), and whether selection importance agrees with allocated recursion depth.

Most important (descending)			Least important (ascending)		
Rank	Gene	Score	Rank	Gene	Score
1	ADAM12	42.58	1	ALS2CR11	6.69
2	IL1R2	42.18	2	SPINK2	6.73
3	SLC6A2	41.96	3	B3GNT3	6.90
4	SRMS	41.93	4	MAST1	7.04
5	MEG3	41.68	5	AGR3	7.05
6	CADM3	41.63	6	PRUNE2	7.22
7	CES1	41.44	7	TRY6	7.26
8	MT1F	40.63	8	MFAP4	7.34

Table 6: Feature importance ranking (marker-query affinity): the most important (descending) and least important (ascending) genes. Each entry is a rank, gene identifier, and importance score.

affinity and mean recursion depth): *which* genes are kept and *how deeply* a kept gene is then iterated are largely independent, complementary signals rather than two views of one quantity, which is consistent with our finding that routing buys compute rather than accuracy. We report this rank correlation transparently rather than as a positive consistency claim. Table 6 shows the most and least important features under this ranking. The full per-gene ranking is emitted to `token_reduction_ranking.csv` and the structured validation record (summary, selected indices, descending/ascending rankings, and selection metadata) to `token_reduction.json`.

Adaptive Recursion Allocates Compute to Driver Genes

Table 7 compares our expert-choice Mixture-of-Recursions router against token-choice routing and the uniform fixed-depth recursion. All three regimes land within about half a macro-F1 point of one another (fixed depth 98.4%, expert-choice 98.7%, token-choice 98.9%), so on this near-saturated task the choice of router does not change accuracy. The value of routing is *compute*, not accuracy: both adaptive routers let most markers exit the funnel early, cutting the mean recursion depth from the full K=4 to 2.75 and spending only 0.60× of the fixed-depth stack FLOPs at no measurable accuracy cost. The depth is not spent uniformly: a small set of genes survives to the deepest step, and their per-cohort recursion depth d_m gives an importance ranking that is intrinsic to the computation rather than read off attention maps. Expert- and token-choice save almost identically here; we adopt expert-choice as the headline because its capacity funnel is load-balanced by construction and needs no auxiliary balancing loss, whereas token-choice can be harder to balance (Bae et al. 2025). We use the gentle capacity funnel $(1, \frac{3}{4}, \frac{1}{2}, \frac{1}{2})$ and full-strength router gates (α=1); an aggressive 0.5^t funnel or a strongly damped gate (α=0.1) starves the *pooled* marker representation and costs several macro-F1 points, an interaction we observed in development.

Routing	Macro-F1	Acc.	Mean depth	Eff. FLOPs	Saving
Expert-choice (ours)	98.7	98.5	2.75	161.5M	0.60×
Token-choice	98.9	98.9	2.69	159.1M	0.59×
Fixed depth (uniform)	98.4	98.5	4.00	268.4M	1.00×

Table 7: Mixture-of-Recursions routing study. Expert-choice (ours) preserves accuracy at a fraction of the fixed-depth compute; "Mean depth" is the average per-gene recursion depth, an intrinsic importance signal.

Recursion	Cohort	Stage	T	N	Depth	Saving
Fixed depth (K=8)	97.4 ± 0.4	45.9 ± 0.5	38.7 ± 0.6	28.8 ± 9.8	8.00	1.00×
Early-exit (K=8, ours)	97.7 ± 0.8	47.3 ± 2.5	36.0 ± 5.1	28.1 ± 3.8	4.75	0.49×

Table 8: Early-exit recursion vs fixed depth at cap K=8 (macro-F1 %, mean ± std over seeds). "Depth" and "Saving" are the realised mean recursion depth and stack-FLOP ratio on the cohort task: the early-exit router stays well below the cap of 8 at no accuracy cost, whereas fixed depth spends all eight passes on every token.

Early-Exit Recursion vs Fixed Depth

The expert-choice router makes the recursion an *early-exit* (early-stopping) process: rather than fixing a depth and running every marker token for all K passes, the router decides which tokens survive each step, so a token's realised recursion depth is learned and most tokens stop early. To make this explicit we set a deep cap K=8 and compare fixed depth (every token runs all eight passes) against the early-exit router (Table 8). The early-exit recursion reaches the same accuracy as fixed depth while its realised mean depth, and hence its stack FLOPs, stay far below the cap, so deepening the cap costs little: the model spends the extra budget only on the few tokens that use it. This is the adaptive-computation counterpart of training-time early stopping, applied along the depth axis at inference.

Marker Selection Study

Does *learning* the markers help, and *how* should one learn them? We compare four selection strategies under an identical token construction and fixed recursion, so only selection differs (Table 9). The two fixed panels (variance, random) see only their chosen genes, a hard drop of all others, while the two differentiable selectors (our cross-attention router and the Concrete relaxation) attend *softly over every gene* during training, so gradients reach the whole transcrip-

Variant	Macro-F1	Acc.	Stack params	FLOPs/cell
Concrete (learned)	99.1	99.2	132,480	268.4M
Router (learned, ours)	98.4	98.5	132,480	268.4M
Variance (heuristic)	97.7	97.9	132,480	268.4M
Random	97.4	97.7	132,480	268.4M

Table 9: Marker-selection study at fixed recursion. Both learned selectors (Concrete and the cross-attention router) beat the variance and random panels; Concrete edges the router here. Margins are small on this near-saturated four-class task (see text). Router and Concrete attend softly over all genes in training and collapse to a hard panel at evaluation; variance and random see only their chosen genes.

tome, and collapse to a hard arg-max panel at evaluation; all four then feed the same fixed-depth recursion, which isolates selection quality. Both learned selectors top the table: the Concrete relaxation reaches 99.1% macro-F1 and our router 98.4%, ahead of the strong variance heuristic (97.7%) and random selection (97.4%). Learning *which* genes are markers therefore helps over fixed heuristics, and the two differentiable selectors are close, with Concrete edging the router here. The margins are small on this near-saturated four-class task: the value of a learned selector is not a large accuracy jump but that it is end-to-end and transfers to settings where a variance prior is uninformative. We keep the cross-attention router as the headline selector because it integrates directly with the recursive marker tokens and the refinement gate, while reporting Concrete as an equally strong differentiable alternative.

Two design choices are needed to make the router learn at all. First, selection must be *soft over all genes*: a hard top-k router only re-ranks genes it has already selected and cannot discover new ones, whereas our router and the Concrete relaxation attend over *every* gene and receive gradient everywhere. Second, the soft selector must be *initialised peaked*, each slot starting on a distinct gene; from a uniform start every marker token is the same average of all genes and the model cannot escape this collapse within a practical budget. In preliminary runs a naive hard router and a uniform-initialised router failed to beat random selection, which motivated both choices; we report this as design rationale rather than a headline result, since those variants are not part of the controlled study above.

Peaked Initialisation Is the Decisive Ingredient

The router relies on two design choices: a *peaked* initialisation that points each query at a distinct gene, and a temperature schedule that anneals the selection softmax from soft to peaked. To isolate their contributions we cross them in a 2×2 grid (peaked vs. uniform initialisation $\times$ annealed vs. constant temperature), each cell averaged over 3 seeds (Table 10). The outcome is unambiguous: peaked initialisation is decisive, lifting macro-F1 from 57.5% to 96.6% (about 39 points), whereas temperature annealing moves it by under one point in either initialisation regime. From a uniform start every marker token is the same average of all genes and the model cannot escape that collapse within the training budget; annealing is a useful refinement on top of a good initialisation, not a substitute for it.

Initialisation	Annealed τ	Constant τ
Peaked (ours)	96.5 ± 1.3	96.6 ± 1.7
Uniform	57.0 ± 5.7	58.0 ± 10.5

Table 10: Initialisation $\times$ annealing ablation for the marker router (macro-F1 %, mean $\pm$ std over 3 seeds). Peaked initialisation is decisive; temperature annealing is a minor refinement on top of it.

Biology-Informed Routing

Section 3 adds a genomap gene-gene-interaction centrality prior to the depth router. We test whether it helps, and whether it is the *real* co-expression structure that matters, by comparing three router priors under otherwise identical training: *none* (the data-only router), *co-expression* (the genomap correlation-graph centrality), and a degree-matched *random graph* control with the same sparsity but shuffled edges (Table 11). We evaluate on the cohort task and the three hard phenotype tasks, since the prior is an empirical-Bayes shrinkage that should help most where the label signal is weak. The comparison of interest is co-expression versus random: a gain there, not merely over *none*, is what would show that biological network structure rather than any additive bias drives the effect.

Finding (reported as a negative result). Across all four tasks the three router priors are statistically indistinguishable: every co-expression cell lies within one standard deviation of both its degree-matched random-graph control and the no-prior baseline (Table 11). On the near-saturated cohort task the prior is mildly negative, as expected once the label signal is already strong. On the weakly-determined phenotype tasks it neither helps nor hurts on the mean, and crucially it does not separate from the random graph, so on these labels we cannot attribute any effect to biological co-expression structure rather than to generic additive bias. The one consistent trend is lower seed-to-seed variance under the co-expression prior on stage and tumour-T, consistent with its intended role as a shrinkage regulariser, but we do not promote a variance effect that does not reach an accuracy gain. We therefore present biology-informed routing as a principled, label-free mechanism whose benefit is *not* realised on the present datasets, and scope its promise to settings with a stronger or more explicitly network-structured signal (Discussion).

Architecture Ablations

Table 12 isolates each component on the full model, reported exactly as we find it. On this four-class task no single component drives accuracy: a single pass ($K{=}1$) nearly matches the four-step model, removing the recursive refinement gate costs under one macro-F1 point, and removing weight sharing changes accuracy by about the same amount

Router prior	Cohort	Stage	T	N
None (baseline)	97.0 ± 1.4	45.7 ± 3.0	40.8 ± 5.1	31.0 ± 3.6
Co-expression (ours)	95.6 ± 0.5	46.8 ± 1.7	40.1 ± 2.4	30.9 ± 5.0
Random graph	95.1 ± 1.3	46.9 ± 3.5	40.4 ± 1.3	31.6 ± 3.2

Table 11: Biology-informed routing ablation (macro-F1 %, mean $\pm$ std over seeds): the genomap gene-gene-interaction centrality prior (co-expression) vs. a degree-matched random-graph control vs. no prior, on the cohort and hard phenotype tasks. The co-expression-vs-random gap isolates the contribution of real biological network structure.

Variant	Macro-F1	Acc.	Stack params	FLOPs/cell
Full model	98.7	98.5	132,992	161.5M
− recursive refinement	97.9	97.7	132,992	161.5M
− recursion (K=1)	98.1	98.1	132,608	67.1M
− weight sharing	98.9	99.0	530,432	161.5M

Table 12: Architecture ablations on the primary head. On this near-saturated task every component moves macro-F1 by under a point and within run-to-run noise; weight-shared recursion is within a fraction of a point of independent layers at a quarter of the stack parameters, so the gains are efficiency, not accuracy.

in the other direction at roughly four times the stack parameters. Every variant sits within the run-to-run noise that the harder-task multi-seed study (Table 13) makes explicit, so we read no ranking into these fractions of a point. The "− weight sharing" variant is exactly a standard K-layer transformer over the M marker tokens, matched in width, depth and token set, so it doubles as our matched-budget standard-transformer baseline. Generic efficient-attention backbones (Linformer, Performer, Nyströmformer (Wang et al. 2020; Choromanski et al. 2021; Xiong et al. 2021)) lower the quadratic attention cost orthogonally to our $\mathcal{O}(M^2)$ marker compression and are a complementary direction we do not benchmark here. The architecture's benefit on this dataset is therefore *efficiency* (weight-shared recursion, adaptive-depth compute saving) and the interpretable recursion-depth signal, rather than an accuracy gain from depth itself; we expect depth to contribute more on harder tasks with finer-grained classes, which we did not test here.

Robust Ablations on Harder Tasks

The cohort task is close to saturated, so all ablations cluster within run-to-run noise and cannot by themselves rank the components. We therefore repeat the ablations on the three genuinely hard phenotype labels (overall stage, tumour T, node N), each over five random seeds, and report mean $\pm$ std macro-F1 (Table 13). We report a negative result here for completeness: on these weakly-determined clinical labels the ablations remain statistically indistinguishable, with all variants, including learned versus random marker selection, falling within overlapping one-standard-deviation bands. The benefit of learned marker selection that is visible on the cohort task (Table 9, router 98.4% vs. random 97.4%) therefore appears to require a clear marker signal in the label; on staging and nodal status, which are only faintly encoded in bulk expression, no selection or recursion choice separates from the others. We accordingly scope the marker-learning claim to tasks with a genuine marker signal rather than asserting it universally. We also sweep the two main knobs on the same hard tasks. The recursion-depth sweep (Table 14) traces the accuracy/compute curve, where accuracy is flat within run-to-run noise across depth while the expert-choice router keeps effective FLOPs below the fixed-depth budget, and the marker-count sweep (Table 15) shows that even M=32 markers nearly match M=512, so the $\mathcal{O}(M^2)$ compression is close to free on these tasks.

Variant	Stage	T	N
Full model	44.5 ± 2.2	39.2 ± 2.8	30.5 ± 2.7
− recursive refinement	43.7 ± 2.6	40.8 ± 2.4	32.0 ± 1.2
− recursion (K=1)	44.4 ± 2.5	38.9 ± 1.4	33.2 ± 1.3
− weight sharing	44.4 ± 1.0	40.0 ± 1.0	32.3 ± 3.5
Fixed depth (uniform)	45.0 ± 2.9	38.9 ± 1.8	28.5 ± 1.8
Token-choice MoR	43.8 ± 1.6	38.6 ± 1.9	32.4 ± 1.9
Variance markers	42.7 ± 2.8	36.1 ± 1.8	28.7 ± 4.4
Random markers	45.2 ± 2.3	40.9 ± 2.7	32.4 ± 2.3

Table 13: Multi-seed ablations (macro-F1 %, mean $\pm$ std over five seeds) on the hard phenotype tasks. The variants overlap within one standard deviation, including learned versus random marker selection: no component is statistically separable on these weakly-determined clinical labels.

K	Eff. FLOPs	Stage	T	N
1	1.00×	45.9 ± 2.4	39.6 ± 1.2	30.1 ± 3.8
2	0.83×	44.7 ± 2.7	40.5 ± 4.1	32.2 ± 3.7
4	0.60×	44.4 ± 5.3	38.7 ± 3.2	29.8 ± 0.9
6	0.53×	45.3 ± 3.2	38.0 ± 1.0	31.9 ± 2.1
8	0.49×	44.0 ± 1.0	38.0 ± 2.1	29.0 ± 4.0

Table 14: Recursion-depth sweep (headline config, three seeds): macro-F1 (%, mean $\pm$ std) and effective FLOPs vs. depth K on the hard tasks.

Generalization to Single-Cell Datasets

Although the controlled study above uses bulk pan-cancer data, SMART makes no assumption about where the expression vector comes from. To test transfer, we run the *identical* headline configuration (cross-attention marker router with expert-choice recursion, unchanged) on four single-cell benchmarks from the genomap capsule (Islam and Xing 2023): Tabula Muris and three curated cross-tissue panels, spanning 10 to 55 cell-type classes and up to 90,579 cells, using each dataset's own train/test split where one is provided. Table 16 reports test accuracy and macro-F1. The same architecture classifies cell types across all four datasets, including the fine-grained 55-class Tabula Muris benchmark, which supports our view that marker-guided recursion is a general mechanism for expression classification rather than something specific to bulk cohorts.

The Pancreas row is the one place where accuracy stays high while macro-F1 drops, and we read it as a property

M	Params	Stage	T	N
32	132,992	42.8 ± 3.5	37.2 ± 0.9	30.0 ± 4.1
64	132,992	43.4 ± 1.7	37.6 ± 1.2	30.6 ± 0.7
128	132,992	44.1 ± 2.1	39.7 ± 1.6	31.8 ± 2.8
256	132,992	44.2 ± 2.7	39.1 ± 4.8	29.1 ± 0.6
512	132,992	44.2 ± 1.5	37.7 ± 0.4	32.9 ± 2.4

Table 15: Marker-count (M) sweep (headline config, three seeds): macro-F1 (%, mean $\pm$ std) and stack parameters vs. the number of marker tokens M.

Dataset	Cells	Genes	Classes	Acc.	Macro-F1
Tabula Muris	54,865	1,089	55	63.2	56.0
Common-class	27,499	1,089	19	83.1	68.7
Prototype	90,579	752	10	93.0	92.7
Pancreas	14,767	1,936	15	87.3	48.2

Table 16: SMART generalises to single-cell cell-type classification: the headline configuration, run unchanged, on four genomap-capsule benchmarks (Islam and Xing 2023) (each dataset's own split where available). Cells, genes and classes vary widely across datasets.

of the input representation rather than a shortcoming of the model. Unlike the other panels, the Pancreas features are flattened 44×44 genomap images (Islam and Xing 2023) rather than a raw named-gene vector, so the cross-attention marker router selects over spatial genomap pixels instead of genes and its inductive bias toward marker genes does not apply; the high overall accuracy then reflects the frequent cell types while the rarer ones are diluted, which is what depresses the macro-average. This is exactly the behaviour we hypothesised for the genomap-derived inputs ahead of the run, and the result confirms that hypothesis. We therefore treat the gene-panel datasets as the on-distribution test of the architecture and keep the genomap-image case as a deliberately out-of-format stress test.

Clinical Phenotype Tasks (genoNet Benchmark)

Beyond cohort of origin, the same tumours carry clinically meaningful phenotype labels. genomap's genoNet classifier (Islam and Xing 2023) reshapes the gene vector into an image and applies a small CNN; we instead keep our model fixed and feed it the raw, full gene vector (all 20,530 genes, 2,738 samples). We run the unchanged headline SMART configuration on the five clinical genoNet tasks (Table 17): three ordinal pathology labels (overall stage, tumour T, node N) and two binary clinical labels (overall survival, tumour status). We deliberately exclude cohort detection from this benchmark: it is near-linearly separable from bulk expression and saturated for any reasonable model, so it is a sanity check rather than a predictive task. SMART degrades gracefully across the genuinely hard staging and survival tasks, which are only weakly determined by bulk expression, matching the difficulty ordering reported for deep classifiers on TCGA RNA-seq (Khalifa et al. 2020; Rukhsar et al. 2022) and for staging in recent multi-omic pipelines (Ghaleb et al. 2025).

Task	Classes	Acc.	Macro-F1	Weighted-F1
Pathologic stage	4	46.0	45.5	45.9
Tumour (T)	4	41.1	38.8	42.0
Node (N)	4	47.4	37.8	49.3
Overall survival	2	71.4	56.4	74.4
Tumour status	2	60.6	53.9	63.8

Table 17: SMART on the five clinical genoNet tasks (unified TCGA table, all genes), run with the unchanged headline configuration. Staging, nodal and survival labels are intrinsically hard from bulk expression.

External Baselines

To position the absolute numbers we compare SMART against strong *nonlinear* tabular learners under the *identical* train/test split and gene set on every genoNet task: a majority-class floor; k-nearest neighbours; an RBF SVM; three forest/boosting ensembles from scikit-learn (random forest, extra trees, histogram gradient boosting); the three widely used gradient-boosting libraries XGBoost, LightGBM and CatBoost; and a one-hidden-layer MLP. We deliberately benchmark against this nonlinear class rather than linear models, which are not informative comparators for an architecture whose claim is parameter efficiency and interpretability rather than raw accuracy. Table 18 reports macro-F1 per task and the across-task mean. For SMART, the model width, marker count, recursion depth and learning rate are selected per task by *validation* macro-F1 over a small grid (selection never sees the test split); the baselines use their standard strong settings. SMART is competitive with these strong gradient-boosted baselines on the hard clinical tasks while additionally yielding an interpretable, compute-allocated marker panel that none of them provide.

We do not retrain large pretrained foundation models (scGPT (Cui et al. 2024), scFoundation (Hao et al. 2024), GET (Fu et al. 2025), network-aware (Zhang et al. 2025), scPlantLLM (Cao et al. 2025)) or the feature-selection-plus-deep-net pipelines that report headline TCGA accuracies (Polepalli 2025; Khalifa et al. 2020; Rukhsar et al. 2022; Kim and Jang 2025; Ghaleb et al. 2025; Rahaman et al. 2025; Shukla, Dwivedi, and Mishra 2025; Bouazza 2025) as head-to-head numbers, because they use different gene panels, splits, augmentation and pretraining corpora and are therefore not directly comparable; we cite them as the broader landscape. Our contribution is orthogonal to that line: parameter efficiency and marker interpretability as an architectural property rather than a property of scale or of an external feature-selection stage.

Router-Based Gene Identification

A central claim is that the architecture *identifies* genes rather than merely classifying. Two intrinsic signals rank genes: the cross-attention router's per-slot selection (which

Method	Stage	T	N	OS	Tumour	Mean
SMART (ours)	**45.5**	**38.8**	**37.8**	**56.4**	**53.9**	**46.5**
Hist GBM	49.9	43.7	34.3	48.5	52.7	45.8
LightGBM	50.7	42.5	36.9	47.3	51.5	45.8
XGBoost	50.2	41.8	34.5	48.5	53.0	45.6
CatBoost	49.1	41.8	32.2	48.3	52.1	44.7
MLP (1x256)	48.5	39.2	34.6	49.8	49.0	44.2
k-NN	48.0	36.3	31.9	47.3	51.3	42.9
Extra Trees	48.7	40.6	31.6	46.0	45.5	42.5
RBF SVM	48.1	40.7	31.5	46.0	45.6	42.4
Random Forest	47.8	38.3	30.9	47.3	44.7	41.8
Majority (floor)	13.0	16.3	17.5	46.0	43.9	27.3

Table 18: SMART vs. strong nonlinear / gradient-boosted baselines (macro-F1, %) on the genoNet tasks under an identical split and gene set; rows sorted by across-task mean. Stage / T / N / OS / Tumour are the five clinical phenotype labels.

Gene	d_m	Gene	d_m
ANKRD2	4.00	*PADI3*	3.98
HAL	4.00	*LGSN*	3.98
IL1A	4.00	*PON1*	3.97
LOC440173	4.00	*ITIH2*	3.97
PLAC1	4.00	*CCL23*	3.95
POMC	3.99	*ROS1*	3.95
CNGB3	3.99	*RBP5*	3.95
GKN2	3.99	*NXPH3*	3.94

Table 19: Top router-identified genes ranked by mean recursion depth d_m (the adaptive compute the model allocates to each). Listed descriptively; the depth ranking is intrinsic to the architecture and needs no post-hoc attribution.

M genes become markers) and each marker's mean recursion depth d_m (how much adaptive computation the model spends on it, Table 7). We treat d_m as a compute-allocation importance score that complements selection and needs no post-hoc attribution.

The deepest-routed genes. Table 19 lists the markers that survive to the deepest recursion steps, ranked by mean depth d_m; these are the genes on which the model spends the most adaptive computation. Several have documented roles in epithelial cancers, for example the receptor tyrosine kinase *ROS1*, an established lung-adenocarcinoma driver, and the inflammatory cytokine *IL1A*. We present this panel *descriptively* rather than as a validation claim: the deepest-routed genes did not all coincide with our small curated marker list, and establishing that the panel is enriched for cancer biology beyond chance requires the systematic test described next, which we leave to future work. The ranking is intrinsic to the architecture and needs no post-hoc attribution.

Stability of the panel across seeds. Because the cohort task is near-saturated and co-expressed markers are redundant, the *identity* of the selected genes is far less stable than the accuracy: across our five seeds the selected marker panels overlap by only a mean Jaccard of 0.03, i.e. different seeds reach the same accuracy through largely disjoint gene sets. We therefore treat a single run's recursion-depth ranking as a per-run, hypothesis-generating signal rather than a fixed biomarker list, and any biological claim should be drawn from the consensus of many runs, not one panel. This instability is itself informative: it quantifies the redundancy of discriminative genes in bulk expression that the marker literature describes qualitatively.

Systematic pathway enrichment. Beyond this curated check, an unbiased enrichment of the full learned panel against Reactome gene sets (Gillespie et al. 2022) (hypergeometric test with FDR control), reported as ranked pathways with effect sizes, is a natural next step that would turn the qualitative marker validation into a quantitative one; we leave it to future work.

5 Discussion and Limitations

SMART shows that biological inductive bias and parameter-efficient recursion can be co-designed: the same mechanism that makes the model small (sharing, compression) also makes it interpretable (markers). Beyond the controlled cohort task, we stress-tested the model on the full $\sim$20,530-gene transcriptome across five clinical phenotype tasks, against ten strong nonlinear baselines, and across four single-cell datasets, which addresses the all-gene regime, the external-baseline comparison, and partial multi-seed variance reporting that an earlier version of this work left open. The headline finding is mixed in an informative way: SMART attains the best across-task mean macro-F1 and wins the binary clinical tasks, but gradient-boosted trees lead on tumour stage and T, so SMART's advantage is efficiency and interpretability at competitive accuracy rather than a universal accuracy gain.

We scope the claims to what the evidence supports, and flag the genuine remaining limitations. (i) *Marker learning is signal-dependent.* The learned router clearly beats random and variance selection on the cohort task, but our five-seed ablations show that on the weakly-determined staging and nodal labels every component, learned selection included, sits within one standard deviation of the others; marker learning helps only where the label carries a marker signal. (ii) *Accuracy ceiling.* Against tuned gradient boosting SMART leads on the mean and on three of five tasks but trails on two; closing the staging gap likely needs ordinal-aware losses or pretraining (Cui et al. 2024; Hao et al. 2024; Fu et al. 2025). (iii) *Generalisation.* Results are single external assembly (TCGA); leave-one-cohort-out and an independent external cohort, plus large-scale pretraining and comparison against efficient-attention backbones (Wang et al. 2020; Choromanski et al. 2021; Xiong et al. 2021), remain future work. (iv) *Routing/refinement interaction and optimisation.* Expert-choice routing under-performs fixed depth unless the funnel is gentle and gates are full-strength, and the differentiable top-k is the main source of optimisation noise; smoother relaxations are a promising direction. (v) *Depth versus statistical prominence.* Recursion depth and a gene's raw expression variance may be partially entangled: a high-variance gene is both easier to select and more likely to be routed deep, so part of the depth signal could reflect statistical prominence rather than biology. We do not disentangle the two here; a rank correlation of d_m against per-gene variance and mean $|z|$ on a held-out split, together with the Reactome enrichment above, is the test needed to establish that depth carries information beyond variance, and we leave it to future work. (vi) *Biology-informed routing is not yet realised.* The genomap interaction prior is principled and label-free, but on our four tasks it does not separate from a degree-matched random-graph control (Sec. 4), so we cannot claim a benefit from biological structure here; testing it where the prior should bite, datasets with stronger or explicitly network-mediated signal, with richer priors (pathway membership, regulatory-network centrality) and the optional logit Laplacian smoothing of Appendix D, is the natural next step. (vii) *Comparisons and profiling left to future work.* We do not yet include a controlled from-scratch transcriptomics-transformer baseline (e.g. a Geneformer- or GexBERT-style stack trained on our split) nor efficient-attention backbones (Wang et al. 2020; Choromanski et al. 2021; Xiong et al. 2021) as head-to-head comparators, and our efficiency evidence is stack FLOPs and training wall-clock rather than end-to-end inference-time and memory profiles across batch sizes; both are clear next steps to substantiate the practical efficiency claim. We report these so the trade-offs are not overstated.

6 Conclusion

We presented SMART, a recursive marker-guided transformer that makes parameter efficiency an architectural property of transcriptomic models. By learning marker genes, compressing around them, and sharing one block across recursive refinement steps, it matches strong cohort-classification accuracy with several times fewer parameters and exposes an interpretable, compute-allocated recursion-depth gene panel. Evaluated on the full transcriptome across five clinical phenotype tasks it achieves the best mean macro-F1 against ten strong nonlinear baselines while remaining far smaller, and it transfers unchanged to single-cell data; our multi-seed study also delineates, honestly, where the marker-learning advantage does and does not hold. We additionally formulate a biology-informed router that folds a label-free genomap gene-gene interaction prior into the recursion decision, with full mathematical and biological grounding; that it does not yet beat a random-graph control on these datasets is reported as an honest negative result and a clear direction for stronger, network-structured settings. The complete pipeline, including all experiments, baselines, and this paper, regenerates from a single command.

References

Aran, D.; Looney, A. P.; Liu, L.; Wu, E.; Fong, V.; Hsu, A.; et al. 2019. Reference-based analysis of lung single-cell sequencing reveals a transitional profibrotic macrophage. *Nature Immunology* 20:163–172.

Bae, S.; Kim, Y.; Bayat, R.; Kim, S.; Ha, J.; et al. 2025. Mixture-of-recursions: Learning dynamic recursive depths for adaptive token-level computation. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Balın, M. F.; Abid, A.; and Zou, J. 2019. Concrete autoencoders: Differentiable feature selection and reconstruction. In *International Conference on Machine Learning (ICML)*.

Bouazza, S. H. 2025. A deep ensemble gene selection and attention-guided classification framework for robust cancer diagnosis from microarray data. *Engineering, Technology & Applied Science Research*.

Cao, G.; Chao, H.; Zheng, W.; Lan, Y.; Lu, K.; Wang, Y.; Chen, M.; Zhang, H.; and Chen, D. 2025. scPlantLLM: A foundation model for exploring single-cell expression atlases in plants. *Genomics, Proteomics & Bioinformatics*.

Choromanski, K.; Likhosherstov, V.; Dohan, D.; Song, X.; Gane, A.; Sarlos, T.; et al. 2021. Rethinking attention with performers. In *International Conference on Learning Representations (ICLR)*.

Cui, H.; Wang, C.; Maan, H.; Pang, K.; Luo, F.; Duan, N.; and Wang, B. 2024. scgpt: Toward building a foundation model for single-cell multi-omics using generative AI. *Nature Methods* 21:1470–1480.

Dehghani, M.; Gouws, S.; Vinyals, O.; Uszkoreit, J.; and Kaiser, L. 2019. Universal transformers. In *International Conference on Learning Representations (ICLR)*.

Franzén, O.; Gan, L.-M.; and Björkegren, J. L. 2019. Panglaodb: A web server for exploration of mouse and human single-cell RNA sequencing data. *Database* 2019:baz046.

Fu, X.; Mo, S.; Buendia, A.; Laurent, A. P.; Shao, A.; et al. 2025. A foundation model of transcription across human cell types. *Nature*.

Ghaleb, M. S.; Al-Berry, M.; Ebied, H. M.; and Tolba, M. F. 2025. A novel statistical feature selection framework for biomarker discovery and cancer classification via multi-omics integration. *BMC Medical Research Methodology*.

Gillespie, M.; Jassal, B.; Stephan, R.; Milacic, M.; Rothfels, K.; Senff-Ribeiro, A.; Griss, J.; et al. 2022. The reactome pathway knowledgebase 2022. *Nucleic Acids Research* 50(D1):D687–D692.

Graves, A. 2016. Adaptive computation time for recurrent neural networks. *arXiv preprint arXiv:1603.08983*.

Hao, M.; Gong, J.; Zeng, X.; Liu, C.; Guo, Y.; Cheng, X.; Wang, T.; Ma, J.; Zhang, X.; and Song, L. 2024. Large-scale foundation model on single-cell transcriptomics. *Nature Methods* 21:1481–1491.

Howlader, K.; Islam, M. T.; and Le, W. 2026. Graph transformer-based pathway embedding for cancer prognosis. *arXiv preprint arXiv:2604.16685*.

Hu, C.; Li, T.; Xu, Y.; Zhang, X.; Li, F.; Bai, J.; Chen, J.; Jiang, W.; Yang, K.; Ou, Q.; Li, X.; Wang, P.; and Zhang, Y. 2023. Cellmarker 2.0: An updated database of manually curated cell markers in human/mouse and web tools based on scRNA-seq data. *Nucleic Acids Research* 51(D1):D870–D876.

Ianevski, A.; Giri, A. K.; and Aittokallio, T. 2022. Fully-automated and ultra-fast cell-type identification using specific marker combinations from single-cell transcriptomic data. *Nature Communications* 13:1246.

Islam, M. T., and Xing, L. 2023. Cartography of genomic interactions enables deep analysis of single-cell expression data. *Nature Communications* 14:679.

Jaber, J., and Jaber, O. 2026. Ouroboros: Dynamic weight generation for recursive transformers via input-conditioned LoRA modulation. *arXiv preprint*.

Jang, E.; Gu, S.; and Poole, B. 2017. Categorical reparameterization with gumbel-softmax. In *International Conference on Learning Representations (ICLR)*.

Jiang, S., and Hassanpour, S. 2025. Transformer-based representation learning for robust gene expression modeling and cancer prognosis. *Scientific Reports* 15.

Khalifa, N. E. M.; Taha, M. H. N.; Ali, D. E.; Słowik, A.; and Hassanien, A. E. 2020. Artificial intelligence technique for gene expression by tumor RNA-seq data: A novel optimized deep learning approach. *IEEE Access* 8.

Kim, J. H., and Jang, J. 2025. Pan-cancer gene set discovery via scRNA-seq for optimal deep learning based downstream tasks. *Scientific Reports*.

Lan, Z.; Chen, M.; Goodman, S.; Gimpel, K.; Sharma, P.; and Soricut, R. 2020. ALBERT: A lite BERT for self-supervised learning of language representations. In *International Conference on Learning Representations (ICLR)*.

Lopez, R.; Regier, J.; Cole, M. B.; Jordan, M. I.; and Yosef, N. 2018. Deep generative modeling for single-cell transcriptomics. *Nature Methods* 15:1053–1058.

Luecken, M. D., and Theis, F. J. 2019. Current best practices in single-cell RNA-seq analysis: A tutorial. *Molecular Systems Biology* 15(6):e8746.

Nouriborji, M.; Rohanian, M.; and Rohanian, O. 2025. Improving recursive transformers with mixture of LoRAs. *arXiv preprint arXiv:2512.12880*.

Polepalli, V. 2025. A novel cVAE-augmented deep learning framework for pan-cancer RNA-seq classification. *arXiv preprint*.

Rahaman, M.; Sarker, B.; Alamin, M. H.; and Ferdousi, F. 2025. An integrated approach for key gene selection and cancer phenotype classification: Improving diagnosis and prediction. *Computers in Biology and Medicine*.

Raposo, D.; Ritter, S.; Richards, B.; Lillicrap, T.; Humphreys, P. C.; and Santoro, A. 2024. Mixture-of-depths: Dynamically allocating compute in transformer-based language models. *arXiv preprint arXiv:2404.02258*.

Regev, A.; Teichmann, S. A.; Lander, E. S.; Amit, I.; Benoist, C.; Birney, E.; et al. 2017. The human cell atlas. *eLife* 6:e27041.

Rukhsar, L.; Bangyal, W. H.; Khan, M. S. A.; Ibrahim, A. A. A.; Nisar, K.; and Rawat, D. B. 2022. Analyzing RNA-seq gene expression data using deep learning approaches for cancer classification. *Applied Sciences* 12(4):1850.

Shazeer, N.; Mirhoseini, A.; Maziarz, K.; Davis, A.; Le, Q.; Hinton, G.; and Dean, J. 2017. Outrageously large neural networks: The sparsely-gated mixture-of-experts layer. In *International Conference on Learning Representations (ICLR)*.

Shen, Z.; Liu, Z.; and Xing, E. P. 2022. Sliced recursive transformer. In *European Conference on Computer Vision (ECCV)*.

Shukla, A. K.; Dwivedi, S.; and Mishra, A. 2025. Discriminative biomarker selection using hybrid multi-population evolutionary computation. *Scientific Reports*.

The Tabula Sapiens Consortium. 2022. The tabula sapiens: A multiple-organ, single-cell transcriptomic atlas of humans. *Science* 376(6594):eabl4896.

Theodoris, C. V.; Xiao, L.; Chopra, A.; Chaffin, M. D.; Al Sayed, Z. R.; Hill, M. C.; Mantineo, H.; Brydon, E. M.; Zeng, Z.; Liu, X. S.; and Ellinor, P. T. 2023. Transfer learning enables predictions in network biology. *Nature* 618:616–624.

Vaswani, A.; Shazeer, N.; Parmar, N.; Uszkoreit, J.; Jones, L.; Gomez, A. N.; Kaiser, L.; and Polosukhin, I. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Wang, S.; Li, B. Z.; Khabsa, M.; Fang, H.; and Ma, H. 2020. Linformer: Self-attention with linear complexity. *arXiv preprint arXiv:2006.04768*.

Weinstein, J. N.; Collisson, E. A.; Mills, G. B.; Shaw, K. R. M.; Ozenberger, B. A.; Ellrott, K.; Shmulevich, I.; Sander, C.; and Stuart, J. M. 2013. The cancer genome atlas pan-cancer analysis project. *Nature Genetics* 45:1113–1120.

Wolf, F. A.; Angerer, P.; and Theis, F. J. 2018. SCANPY: Large-scale single-cell gene expression data analysis. *Genome Biology* 19:15.

Xiong, Y.; Zeng, Z.; Chakraborty, R.; Tan, M.; Fung, G.; Li, Y.; and Singh, V. 2021. Nyströmformer: A nyström-based algorithm for approximating self-attention. In *AAAI Conference on Artificial Intelligence*.

Yang, F.; Wang, W.; Wang, F.; Fang, Y.; Tang, D.; Huang, J.; Lu, H.; and Yao, J. 2022. scbert as a large-scale pretrained deep language model for cell type annotation of single-cell RNA-seq data. *Nature Machine Intelligence* 4:852–866.

Zhang, M.; Swamy, V.; Dupire, L.; Cassius, R.; Kanatsoulis, C.; Paull, E.; Karaletsos, T.; and Califano, A. 2025. A cellular network-aware foundation model improves single-cell level predictions. In *Clinical Cancer Research (AACR AI/ML Special Conf.)*.

Zhou, X.; Huang, H.; Wang, Z.; and He, R. 2024. RISTRA: Recursive image super-resolution transformer with relativistic assessment. *IEEE Transactions on Multimedia*.

A Dataset Details

We use three data sources, summarised in Table 20. All gene expression is bulk or single-cell RNA-seq; bulk cohorts are Illumina HiSeqV2 $\log_2(\text{norm\_count} + 1)$ profiles pulled from the UCSC Xena hub, and single-cell sets are the genomap capsule datasets converted to plain CSV.

TCGA pan-cancer bulk (4 cohorts). The primary classification benchmark of the main paper. Four cohorts (breast/BRCA, head-and-neck/HNSC, lung/LUNG, thyroid/THCA) are concatenated; per-gene z-scoring is fit on the training split and the top 4000 high-variance genes of the $\sim$20,530 measured are retained. The label is cohort of origin.

Unified BIO5 phenotype table (genoNet benchmark). The same tumours assembled into one table of 2,738 samples $\times$ 20,530 genes with six aligned phenotype labels (Sec. B). The full gene vector (no high-variance pre-filter) is used, so this is the "all data" setting for SMART and for every external baseline.

Single-cell genomap datasets. Four datasets from the genomap capsule (Islam and Xing 2023), converted to readable CSV (expression + labels + split where provided): Tabula Muris (a fine-grained 55 cell-type panel), a 19-class common-class panel, a 10-class prototype panel, and a 15-class pancreas panel whose features are flattened 44×44 genomap images rather than a raw gene vector.

Dataset	Samples	Features	Labels	Source
TCGA pan-cancer (4 cohorts)	3,485	4,000/20,530	4 cohorts	bulk RNA-seq (Xena)
Unified BIO5 (genoNet)	2,738	20,530	5 tasks	bulk RNA-seq (Xena)
Tabula Muris	54,865	1,089	55 cell types	single-cell (genomap)
Common-class	27,499	1,089	19 cell types	single-cell (genomap)
Prototype	90,579	752	10 cell types	single-cell (genomap)
Pancreas	14,767	1,936	15 cell types	single-cell (genomap)

Table 20: All datasets used. Features for the bulk cohort task are the top high-variance genes; the genoNet table and single-cell sets use their full feature set. Counts are read directly from the data.

B Task Descriptions

The genoNet benchmark contains five clinical classification tasks defined on the unified BIO5 table; each is described below and their class distributions are listed in Table 21. All tasks share the same samples and gene set and use the identical stratified 80/20 split, so they differ only in the target label. (Cohort-of-origin detection is excluded: it is near-linearly separable from bulk expression and saturated, so it is a sanity check rather than a predictive task; it remains the main-paper four-cohort result.)

- **Pathologic stage** – the overall AJCC stage grouped into four ordered levels (Stage I–IV). Only weakly determined by bulk expression.
- **Primary tumour (T)** – size/extent of the primary tumour on the four-level TNM T-axis (T1–T4).
- **Regional lymph node (N)** – nodal involvement on the TNM N-axis (N0–N3); strongly imbalanced, with N3 very rare.
- **Overall-survival status** – a binary survival label; the majority class dominates, so macro-F1 (not accuracy) is the meaningful metric.
- **Tumour status at follow-up** – binary tumour-free vs. with-tumour at last follow-up; also imbalanced.

The single-cell datasets each define one cell-type classification task with the number of classes given in Table 20; we run SMART on them unchanged to test transfer beyond bulk data.

Task	Type	Class distribution (count)
Pathologic stage	4-way ordinal	Stage I: 966, Stage II: 886, Stage III: 528, Stage IV: 358
Primary tumour (T)	4-way ordinal	T1: 700, T2: 1,323, T3: 458, T4: 257
Regional lymph node (N)	4-way ordinal	N0: 1,474, N1: 827, N2: 380, N3: 57
Overall-survival status	binary	class 0: 407, class 1: 2,331
Tumour status at follow-up	binary	class 0: 2,139, class 1: 599

Table 21: Per-task class distributions for the five genoNet phenotype tasks (counts over all 2,738 samples), read directly from the data. The staging, node and clinical tasks are markedly imbalanced.

C Effective-FLOPs Accounting

The compute numbers report per-sample FLOPs *of the recursive transformer stack*, the only component the routing study changes. One application of the shared block to a tokens costs $\phi(a) = 4a^2 d + 4\,a\,d\,d_{\mathrm{ff}}$, the first term the multi-head self-attention ($\mathcal{O}(a^2 d)$ query/key/value and output projections plus the score-value products) and the second the two-layer feed-forward network ($\mathcal{O}(a\,d\,d_{\mathrm{ff}})$); we count these matmuls and treat LayerNorm, the lightweight router head and the final classifier as lower-order. The *nominal* fixed-depth cost is K blocks over all M markers, $\Phi_{\mathrm{nom}} = K\,\phi(M)$. The *effective* cost sums one block over the tokens actually active at each step, $\Phi_{\mathrm{eff}} = \sum_{t=1}^{K} \phi(a_t)$, where a_t is the mean number of markers the expert-choice funnel keeps active at step t, measured on the test set (so depth aggregation is over the empirical per-step survivor counts, not a nominal schedule). The saving ratio reported in the routing study is $\Phi_{\mathrm{eff}}/\Phi_{\mathrm{nom}}$. Gene embedding and marker selection are $\mathcal{O}(Nd)$, run once before the stack and are identical across all routing modes, so they are excluded from this stack-level comparison; total parameter counts (Table 4) do include them.

D Theoretical Foundation of the Router

This appendix gives the mathematical and biological grounding for SMART's router, both the data-driven Mixture-of-Recursions core and the biology-informed prior of Sec. 3.

Routing as conditional computation. The router implements *conditional computation*: rather than sending every token through a fixed stack, a learned policy routes each token to a token-specific amount of compute. This is the gating principle of sparsely-gated mixtures of experts (Shazeer et al. 2017), reused by Mixture-of-Depths (Raposo et al. 2024) and Mixture-of-Recursions (Bae et al. 2025); the "experts" here are recursion *depths* of one shared block rather than separate sub-networks, which is what couples adaptive computation (Graves 2016) to weight sharing.

Token-choice math. Let $\mathbf{h}_m \in \mathbb{R}^d$ be a marker token and $\mathbf{W}_r \in \mathbb{R}^{K\times d}$ the router. The token is scored over the K candidate depths, softmaxed, and assigned the arg-max depth:

$$\mathbf{z}_m = \tfrac{1}{\tau}\mathbf{W}_r\mathbf{h}_m, \quad \mathbf{p}_m = \alpha\,\mathrm{softmax}(\mathbf{z}_m), \quad i_m = \arg\max_i p_{m,i}, \tag{3}$$

so token m rides the shared block through steps $0,\dots,i_m$ and exits. The realised depth is $d_m = i_m{+}1$.

Expert-choice math. Here a per-step scalar router scores the currently active tokens and a capacity c_t keeps the top-$\lceil c_t M\rceil$; survivors are the only candidates at the next step. A token's depth d_m is the deepest step it survived. Expert-choice is load-balanced by construction (each step keeps a fixed fraction), so it needs no balancing loss; token-choice does not balance itself and adds the Switch term $\mathcal{L}_{\mathrm{bal}} = K\sum_i P_i f_i$ (mean routing probability P_i times dispatch fraction f_i at depth i) (Shazeer et al. 2017). Both add the z-loss $\mathcal{L}_z = \mathbb{E}[(\mathrm{logsumexp}\,\mathbf{z})^2]$ to keep logits bounded.

The differentiable handle. The discrete choice ($\arg\max$ / top-k) has zero gradient almost everywhere, so a bare router could not learn. SMART, like MoR, keeps the soft probability of the chosen route as a multiplicative *gate* on the block output, $\mathbf{o}_m = g_m\, f_\theta(\mathbf{h}_m)$ with $g_m = \sigma(\tilde{r}_m)$ (or p_{m,i_m}). Because g_m is smooth in $\mathbf{W}_r$, the chain $\mathcal{L} \leftarrow \mathbf{o}_m \leftarrow g_m \leftarrow \mathbf{W}_r$ is unbroken: the hard choice does the routing, the soft gate carries the gradient (a straight-through-style estimator). The biological prior of Eq. (1) is an additive constant in this logit, so it shifts the decision without breaking this path.

Biological foundation of the prior. Three decades of single-cell biology rest on two facts the prior encodes. (i) A small set of *marker* genes carries most discriminative signal (Ianevski, Giri, and Aittokallio 2022; Franzén, Gan, and Björkegren 2019; Hu et al. 2023), so compute should be unevenly allocated. (ii) Gene regulatory and protein-interaction networks are approximately scale-free: a few high-degree *hub* genes (master regulators) exert outsized influence, and perturbing a hub propagates across the network. We operationalise "hub" as eigenvector centrality on the genomap co-expression graph $\mathbf{W}$: the leading eigenvector $\mathbf{v}$ of $\mathbf{W}\mathbf{v} = \lambda\mathbf{v}$ scores each gene by the centrality of its neighbours, recursively, so a gene is central if it co-expresses with other central genes. We z-score $\mathbf{v}$ to form π. The prior thus tells the router, before any training, which genes are network hubs worth deeper computation.

Empirical-Bayes reading. Equation (1) is a log-linear prior on the routing decision: $\tilde{r}_m^{(t)} =$ (data evidence)$+\beta_t\,\pi_m$ is the unnormalised log-score of keeping token m, with $\beta_t\pi_m$ a Gaussian-like prior mean. The anneal $\beta_t = \beta_0(1 -$ progress) is shrinkage whose strength decays as data evidence accumulates, the standard empirical-Bayes behaviour: prior-dominated when the likelihood is uninformative (early training, random hidden states), data-dominated later. This is most defensible exactly in the low-signal regime (stage, node), where the likelihood alone barely separates the depths.

Leakage safety. $\mathbf{W}$ is computed from training-split *expression only*; no phenotype label enters it. The prior therefore injects network topology, not the answer, which is why a marker-recovery or gene-discovery claim under this prior is not circular, unlike a prior built from curated cohort-specific marker lists.

Optional pathway-graph smoothing. The additive bias treats genes independently. Co-pathway genes should route coherently, which one obtains by smoothing the logits over $\mathbf{W}$ with the normalised Laplacian $\mathbf{L} = \mathbf{I} - \mathbf{D}^{-1/2}\mathbf{W}\mathbf{D}^{-1/2}$, $\hat{\mathbf{r}}^{(t)} = \tilde{\mathbf{r}}^{(t)} - \gamma\,\mathbf{L}\tilde{\mathbf{r}}^{(t)}$, i.e. graph-Laplacian (label-propagation) regularisation: a gene borrows routing strength from its network neighbours. This is one sparse matrix-vector product per step and adds no transformer parameters, so the parameter-efficiency claim is untouched; we expose it as an option and leave its evaluation to future work.